

PAPER_I_EXT_AUDIT — 04_lean_certificate

Script 04 — Lean Formal Certificate Audit (Gate H)

Audit: PAPER_I_EXT_AUDIT · **Script:** 04_lean_certificate.py

Toolchain: Lean 4 v4.28.0 + Mathlib v4.28.0 (elan 4.2.3, lake 5.0.0)

Purpose

Independently verify the manuscript's formal-verification claims for Theorem 1.1 (paperI_main) and Proposition B.1 (residual_duality).

Independent build

The release Lean sources were copied to a ****fresh** directory with no inherited .lake build** (C:\Users\jtrav\l), the Mathlib olean cache fetched via lake exe cache get (community Azure cache; not the author's build), and lake build PaperI.PaperI_Statement run to completion. The build log and script are archived as independent_build.log / independent_build.sh; the axiom lines as independent_axioms_and_exit.txt. (Windows long-path handling required a short build root + core.longpaths=true via GIT_CONFIG_* env; see ENVIRONMENT.md.)

Result: Build completed successfully (8027 jobs), **exit 0**.

PaperI.paperI_main depends on axioms: [propext, Classical.choice, Quot.sound]

PaperI.Split.residual_duality depends on axioms: [propext, Classical.choice, Quot.sound]

Every one of the ~30 intermediate declarations likewise depends only on the three standard foundational axioms. No sorryAx, no native_decide, no added axiom.

Sub-checks

Check	Result
H1 paperI_main : $\Phi \leq n^2/6 + n$, $\Phi = \text{edgeCount} - 2 \cdot \text{nu3star}$ (= Theorem 1.1)	PASS
H2 sources free of real sorry/admit/native_decide/axiom (comments excluded)	PASS (5/5 files clean)
H3 SHA-256 of released lean tree == 04_integrity/SHA256SUMS.txt	PASS (8/8 files MATCH)
H4 independent fresh-tree build: exit 0; axioms [propext, Classical.choice, Quot.sound]; no sorryAx	PASS
H5 independent axioms == recorded gate_logs/print_axioms_residual_and_main.log	PASS (identical)

Verdict

GATE H: PASS. The Lean statement is exactly Theorem 1.1; the released sources are sorry-/axiom-clean and match the recorded manifest by SHA-256; and an ****independent rebuild**** reproduces Build completed successfully with paperI_main (and the LP-duality node residual_duality) depending only on Lean/Mathlib's three standard foundational axioms. The finite LP strong duality used in Prop B.1 is derived inside the development (residual_duality $\leftarrow$ covering_packing_duality $\leftarrow$ finite Farkas), not postulated as a project axiom, consistent with the manuscript's Appendix C.